

Models comparison

Concept Overview

A model architecture is going to affect the battery consumption, spatial localization accuracy and localized thermal output.

Two choices:

1. [CNNs](#) (Convolutional Neural Networks)
 2. [Transformers](#) (Vision Transformers / ViTs)
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CNN Architectures

A convolutional neural network uses convolutional layers that process images as spatial data. Are efficient because they exploit the fact that image pixels have local spatial structure, reducing the number of parameters compared to fully connected networks.

- Efficient on edge hardware
- preserve spatial information
- work well for bounding boxes
- many pretrained models
- can run in real time

1. The YOLO (You Only Look Once) Family

Is a real-time object detection algorithm that processes images in a single forward pass, making it significantly faster than two-stage detectors.

The algorithm divides images into grids, predicts bounding boxes with confidence scores, and uses Non-Maximum Suppression to filter overlapping detections.

YOLO reaches more than twice the mean Average Precision (mAP) compared to other real-time systems, which makes it a great candidate for real-time processing.

- has pretrained models
- YOLO-Nano based on design principles from the YOLO family of single-shot object detection network architectures, is coupled with machine-driven design exploration to create a compact network with highly customized module-level macroarchitecture and microarchitecture designs tailored for the task of embedded object detection.
- yolo detection model
- nano - fastest, accuracy lowest but enough
- small - fast, accuracy better than nano

1. YOLO26n

- edge-optimized model
- supporting five core tasks: object detection, instance segmentation, pose estimation, oriented object detection (OBB), and image classification.

2. YOLO11n

- Developed by Ultralytics
- supports multiple computer vision tasks: object detection, instance segmentation, image classification, keypoint detection, and oriented bounding box (OBB) detection

3. YOLOv10n

- elimination of Non-Maximum Suppression (NMS) during inference, which traditionally added latency to the detection pipeline.
- introduces consistent dual assignments for NMS-free training, combining one-to-many and one-to-one label assignments
- YOLOv10-S is 1.8× faster than RT-DETR-R18 with similar accuracy, while using 2.8× fewer parameters

4. YOLOv8n

- Introduces a more modular and flexible design, allowing easier customization and fine-tuning.
- optimize speed and accuracy trade-offs, with smaller model sizes aimed at real-time applications on edge devices
- It adds support for custom data to integrate easily with custom datasets, making it versatile for specific applications.

5. YOLO12n

- introduces attention mechanisms
- improving the model's ability to focus on relevant image regions
- version achieves better accuracy on complex scenes with overlapping objects while maintaining real-time performance

2. Mobile-First & Compound Scaled CNNs

- [MobileNet SSD \(v2, v3, & v4\)](#)

- combines the lightweight architecture of MobileNet with the single-shot detection mechanism of SSD
- uses depthwise separable convolutions to reduce computation
- good for:
 - mobile deployment
 - TensorFlow Lite
 - simple object detection
- may have lower accuracy than YOLO

- [EfficientDet-Lite \(Lite0 through Lite4\)](#)

- Google's mobile framework

- uses compound scaling and a Bidirectional Feature Pyramid Network (BiFPN) to optimize feature reuse, to balance speed and accuracy
 - good for:
 - TensorFlow Lite
 - mobile/embedded inference
 - [Edge Impulse FOMO](#)
 - ultra-lightweight
 - designed for constrained embedded systems
 - it predicts object location on a grid heatmap
 - rough object location
 - simple tracking
 - lets you count multiple objects and find their location in an image in real-time using up to 30x less processing power and memory than MobileNet SSD or YOLOv5.
 - if you need to know the exact location of something, or if you want to count multiple things (the simple classifier cannot do that) - but it's computationally much more intensive, and you typically need much more data for it.
 - cuts off the last layers of a standard image classification model and replaces this layer with a per-region class probability map
 - gives you a heatmap of where the objects are.
 - [FastestDet](#)
 - ultra-lightweight, single-stage detector for ultra-low latency scenarios
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Transformer & Hybrid Architectures

Transformers can model global relationships in an image better than pure CNNs. This helps in complex scenes, occlusion and unusual viewpoints. May be harder to run on embedded hardware.

1. Pure Edge DETection TRansformers (DETRs)

- [RF-DETR-Nano](#)
 - developed by Roboflow's
 - edge-first transformer that uses DINOv2 vision transformer backbone and that supports object detection, segmentation and keypoint detection
- [Mobile-DETR](#)
 - Specifically engineered for edge mobile processors.

2. Hybrid Convolution-Transformer Networks

Combine CNN efficiency with transformer global reasoning.

- [MobileViT \(v1, v2, & v3\)](#) \
 - combines MobileNet's local convolution processing with transformers-style global context
 - designed for mobile vision tasks
 - It replaces standard convolutions with global processing blocks to limit parameters.
 - good for:
 - mobile image classification
 - [EdgeViT](#)
 - uses efficient local-global-local information exchange mechanism to reduce the computational cost of self-attention.
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Deployment and Export Formats

TensorFlow Lite

- deploy trained machine learning models directly on edge devices such as mobile phones, embedded systems and IoT devices without relying on cloud resources

ONNX

- **Open Neural Network Exchange** is an open source ai ecosystem of technology companies and research organizations that establish open standards for representing machine learning algorithms and software tools to enable a standard format for representing machine learning models.
- Enable developers to move machine learning models between different frameworks

NCNN

- is a high-performance neural network inference framework optimized for mobile, embedded, and desktop deployment
- Developers can deploy deep learning models efficiently on phones, PCs, browsers, and edge devices

Edge Impulse

- is an end-to-end edge AI platform that simplifies embedded machine learning development from data collection and labeling to model training, optimization, validation, and deployment.

Links

[CS231n Deep Learning for Computer Vision](#)

[A Comprehensive Review of YOLO Architectures in Computer Vision: From YOLOv1 to](#)

[YOLOv8 and YOLO-NAS](#)

[ultralytics/ultralytics: Ultralytics YOLO26, YOLO11, YOLOv8 — object detection, instance segmentation, semantic segmentation, image classification, pose estimation, object tracking](#)
[Explore Ultralytics YOLOv8 | Ultralytics](#)

[\[1910.01271\] YOLO Nano: a Highly Compact You Only Look Once Convolutional Neural Network for Object Detection](#)

[STMicroelectronics/yolov8n · Hugging Face](#)

[THU-MIG/yolov10: YOLOv10: Real-Time End-to-End Object Detection \[NeurIPS 2024\]](#)

[YOLO Object Detection Explained: A Beginner's Guide | DataCamp](#)

[MobileNet SSD in PyTorch: A Comprehensive Guide — codegenes.net](#)
[IJNRD2407327.pdf](#)

[FOMO - Edge Impulse Documentation](#)

[Edge-FastestDet: a real-time and ultra-lightweight model for detection](#)

[roboflow/rf-detr: RF-DETR is a real-time object detection and segmentation model architecture developed by Roboflow, SOTA on COCO, designed for fine-tuning. \[ICLR 2026\]](#)

[RF-DETR — Real-Time Object Detection & Segmentation by Roboflow](#)

[MobileViT · Hugging Face](#)

[EdgeViT: Efficient Vision Transformers for Edge](#)

[onnx/docs/Overview.md at main · onnx/onnx](#)

[Open Neural Network Exchange - Wikipedia](#)

[Tencent/ncnn: ncnn is a high-performance neural network inference framework optimized for the mobile platform](#)

[Edge Impulse - The Leading Edge AI Platform](#)

[Edge Impulse - STMicroelectronics](#)

[Introduction to TensorFlow Lite - GeeksforGeeks](#)